Wasuthorn Ausrivong



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EDUCATION

Graduate School of Engineering Science, **Osaka, Japan**
Osaka University **October 1, 2025 - Now**
Master's degree, Department of Systems Innovation,
Division of Systems Science and Applied Informatics, Area of System Theory.
Robot Mechanics Group.

Institute of Field Robotics, Bangkok, Thailand
King Mongkut's University of Technology Thonburi August 7, 2020
Bachelor of Robotics and Automation Engineering (link), GPA: 3.46 (link)
(Second-class honors, ranked 9th in the class of the year),
Senior Project Topic: *Learning to Adjust Locomotion for Damaging Walking Robot*

Bangkok Christian College Bangkok, Thailand
High School Diploma (Math and Science Program), GPA: 3.43 2015

RESEARCH AREAS

Climbing Robots • Industrial Inspection Robots • Autonomous Locomotion • Modular Robots • Bio-inspired Robots • Field Robotics

SKILLS

- **Programming & Software:** C++ • C • Python • SolidWorks • ROS • Arduino • SQL • Git • Linux (Ubuntu) • MATLAB.
- **Research & Innovation:** Experimental Robotics • Scientific Writing • Data Analysis • Technology Transfer • Patent Development.
- **Industrial Applications:** Inspection Robotics • Magnetic Wall-Climbing Robots • Furnace Cleaning Robots • Non-Destructive Testing (VT, MT).
- **Professional Skills:** Proposal Development • Budget Management • Technical Leadership • Project Management • Cross-functional Collaboration • Technical Presentation • Engineering Research Mentorship

Language Skills

- Thai: Excellent (Native Speaker)
- English: IELTs Overall Band score: 6.0, CEFR Level: B2. (link)

SCHOLARSHIP & AWARD

- Honorary Awards, given by Vidyasirimedhi Institute of Science and Technology (VISTEC) to researchers who publish papers in top-tier international conferences in Computer Science and Information Technology, as recognized by CSRankings or ranked A* by CORE. 2026
- VISTEC Honors Awards, given by Vidyasirimedhi Institute of Science and Technology (VISTEC) for outstanding achievements that brought recognition to the institute. (link) 2025
- Japanese Government (Monbukagakusho, MEXT) Scholarship, master's degree, The Graduate School of Engineering Science, Osaka University. (link) 2025
- Delta Internship 2019 Scholarship awarded by Delta Electronics company to students with outstanding research topics. (link) 2019
- Students Internship for Overseas Research Collaboration scholarship given by Institute of Field Robotics to students doing the internship and research aboard. (link) 2019
- Enhancing Student Mobility scholarship given by the International Affairs office of King Mongkut's University of Technology Thonburi to students doing the internship and research aboard. 2019
- Academic Excellence Scholarship awarded by the Institute of Field Robotics to students ranked in the top 5 of the semesters. 2016

WORK EXPERIENCE

Senior Robotics Research Engineer	Vidyasirimedhi Institute of Science and Technology (VISTEC), School of Information Science and Technology (IST), Thailand	JUL 2021 – SEP 2025

- Efficiency ImpRovEment through FIred Heater/FurNace ClEaning (REFINE project), (link)
- Advanced Hybrid Legged-Wheeled Robot for Internal Vessel Inspection (HERO project), (link)
- AdVanced Pipe Inspection Robot System (AVIS project), (link)

Business Analyst (IT)	Kasikorn Business Technology Group (KBTG), Thailand	AUG 2020 – JUN 2021
Teacher Assistant (TA)	Institute of FIeld roBOtics (FIBO), King Mongkut's University of Technology Thonburi (KMUTT), Thailand	AUG 2019 – JAN 2020

- Basic Control Theory and Intro To Robotics subjects.

Internship	The NEUTRON Laboratory, Nanjing University of Aeronautics and Astronautics, China.	JUL 2019 - AUG 2019

PATENT

- Manoonpong, P., Terapongnivat, K., Janna, R., Nantareekurn, W., Ausrivong, W., Srisuchinnawong, A., Pewkliang, S. ***Wheel-legged Robot***, Innovation Patent, PCT International Application No. PCT/IB2025/056485, Filing Date: 26 June 2025. (link)
- Manoonpong, P., Srisuchinnawong, A., Phongaksorn, K., Ausrivong, W., Kosanant, P., Kasemwarapach, T. ***A Method and System for Controlling Motion of a Climbing Robot***, Innovation Patent, PCT International Application No. PCT/IB2021/060011, Filing Date: 29 October 2021. (link)

PUBLICATION

2026	Nontanut Chitsiriparat, Run Janna, Theerawath Phetpoon, Wasuthorn Ausrivong, Kanut Tarapongnivat, Worameth Nantareekurn, Suppachai Pewkliang, and Poramate Manoonpong, ***"Automated Weld Seam Detection and 3D Map Integration for Industrial Vessel Inspection Using Depth Cameras and 3D LiDAR***," 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, Pennsylvania, 2026. (link, video, poster, graphical abstract)	IROS Conference
2026	W. Ausrivong, S. Tadokoro, K. Oguma, Y. Okada, K. Ohno, S. Miyake, K. Abe, M. Watanabe and K. Tadakuma. ***"Magnetic Spherical Omnidirectional Wheel Mechanism".*** Robotics and Mechatronics Conference 2026 (ROBOMECH2026), Fukuoka, Japan, 2026. (link., video, graphical abstract.)	ROBOMECH Conference
2026	Wasuthorn Ausrivong, Masahiro Watanabe, and Kenjiro Tadakuma. ***"Development of an Omnidirectional Magnetic Climbing Robot for Thin Ferrous and Non-Ferrous Surfaces".*** Robotics and Mechatronics Conference 2026 (ROBOMECH2026), Fukuoka, Japan, 2026. (link, video, graphical abstract.)	ROBOMECH Conference
2026	Wasuthorn Ausrivong, Arthicha Srisuchinnawong, and Poramate Manoonpong. ***"Inchworm-inspired Adaptive Multimodal Neural Control for an Autonomous Inspection Robot".*** 2026 IEEE International Conference on Robotics & Automation (ICRA), Vienna, Austria, 2026. (video1, video2, poster, graphical abstract, slides.)	ICRA Conference
2025	Sanpoom Punapanont, Thipawan Pairam, Wasuthorn Ausrivong and Poramate Manoonpong, ***"REFINE-bot: Furnace Cleaning Robot for Heat-transfer Efficiency Improvement,"*** 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, 2025, pp. 15641-15648, doi: 10.1109/IROS60139.2025.11247011. (link, video, poster, graphical abstract, slides.)	IROS Conference
2023	Arthicha Srisuchinnawong, Kitti Phongaksorn, Wasuthorn Ausrivong, and Poramate Manoonpong. (2023) ***Adaptive Bipedal Robot Walking on Industrial Pipes under Neural Multimodal Locomotion Control: Toward Robotic Out-pipe Inspection.*** IEEE/ASME Transactions on Mechatronics. DOT: 10.1109/TMECH.2023.3293950. (link, video.)	Trans. Mechatronics Journal
2023	Kanut Tarapongnivat, Run Janna, Worameth Nantareekurn, Wasuthorn Ausrivong, Arthicha Srisuchinnawong, Naris Asawalertsak, Suppachai Pewkliang, and Poramate Manoonpong. (2023) ***Hybrid Omnidirectional Wheeled Climbing Robot with an Electromagnet for Inspection***, the 26th issue of the International Conference Series on Climbing and Walking Robots and the Support Technologies for Mobile Machines (CLAWAR). (link, video.)	CLAWAR Conference
2023	Thirawat Chuthong, Wasuthorn Ausrivong, Binggwong Leung, Jettanan Homchanthanakul, Nopparada Mingchinda, Poramate Manoonpong. "**Nature Inspired Machine Intelligence**	Standalone Videos

from Animals to Robots". Standalone Videos presented at the 2023 IEEE International Conference on Robotics and Automation (ICRA). (video.)

PROJECTS

Jan 2024 – Dec 2025	**Efficiency ImpRovEment through FIred Heater/FurNace ClEaning (REFINE project)**	VISTEC
	› Senior Robotics Research Engineer, Project website: link.	
Jan 2023 – Dec 2025	**Advanced Hybrid Legged-Wheeled Robot for Internal Vessel Inspection (HERO project)**	VISTEC
	› Senior Robotics Research Engineer, Project website: link.	
Jul 2021 – Dec 2025	**AdVanced Pipe Inspection Robot System (AVIS project)**	VISTEC
	› Robotics Research Engineer, Project website: link.	
Jan 2019 – May 2019	**Learning to Adjust Locomotion for Damaged Walking Robot (Senior-Thesis)**	FIBO
	› Developed and implemented a Model-based Meta-Reinforcement Learning algorithm to adapt locomotion behavior when the robot's legs were damaged.	
	› Researched and evaluated the capabilities of the new behavior using the Mujoco physics engine simulation in Linux environments.	
	› Slide: link.	
Aug 2019– Dec 2019	**Manufacturing and Automation Class Project**	FIBO
	› Designed and implemented communication between a computer and a Programmable Logic Controller (PLC) for the UR5 robotic arm in real robot applications, including preliminary simulations for overall process automation.	
Jun 2019 – Jul 2019	**Q-learning for Walking Robot (Internship)**	Neutron Laboratory at Nanjing University of Aeronautics and Astronautics
	› Designed and implemented a Q-learning algorithm for Lilibot (a 4-legged robot) to enable walking behavior in simulation using the ROS framework and V-REP simulation in Linux environments.	
	› Slide: link.	
Jan 2019 – May 2019	**Parcel Handling Robot with Palletizing Robot**	FIBO
	› Developed and implemented Forward and Inverse Kinematics, Forward and Inverse Dynamics, and robot control for a palletizing robot.	
	› Designed and executed Motion Planning, Trajectory Planning, and preliminary simulations for pick-and-place tasks.	
	› Slide: link., Video: Video-1, Video-2, Video-3	
Aug 2018– Dec 2018	**Chicken packaging Picker Project with Cartesian robot**	FIBO
	› Designed and built the structure of a Cartesian robot from scratch and fine-tuning the PID control system.	
	› Developed communication protocols between a computer and a microcontroller for control purposes.	

LEADERSHIP

2016 - 2020	**Board member of FIBO Student Association**	FIBO
	› Served as an Election Commissioner for the FIBO Student Association in 2016 and 2017	
2015 - 2020	**Class President, Field Robotics and Automation Bachelor's Degree (FRAB #3)**	FIBO
	› Vice-president of *FIBO ROBOCAMP* 6 and *FIBOLIAN Camp* in 2017 and 2018	

ACTIVITIES

- AI and Robotics innovation Contest Hackathon (AIROBIC 2018)
- Global Project Based Learning 2017(GPBL)
- SCG Innovation Challenge 2016